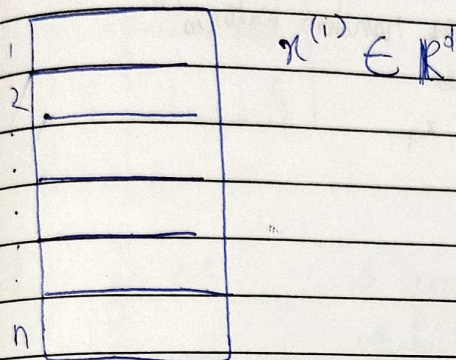


* Random Forest :-

$X = N$ samples

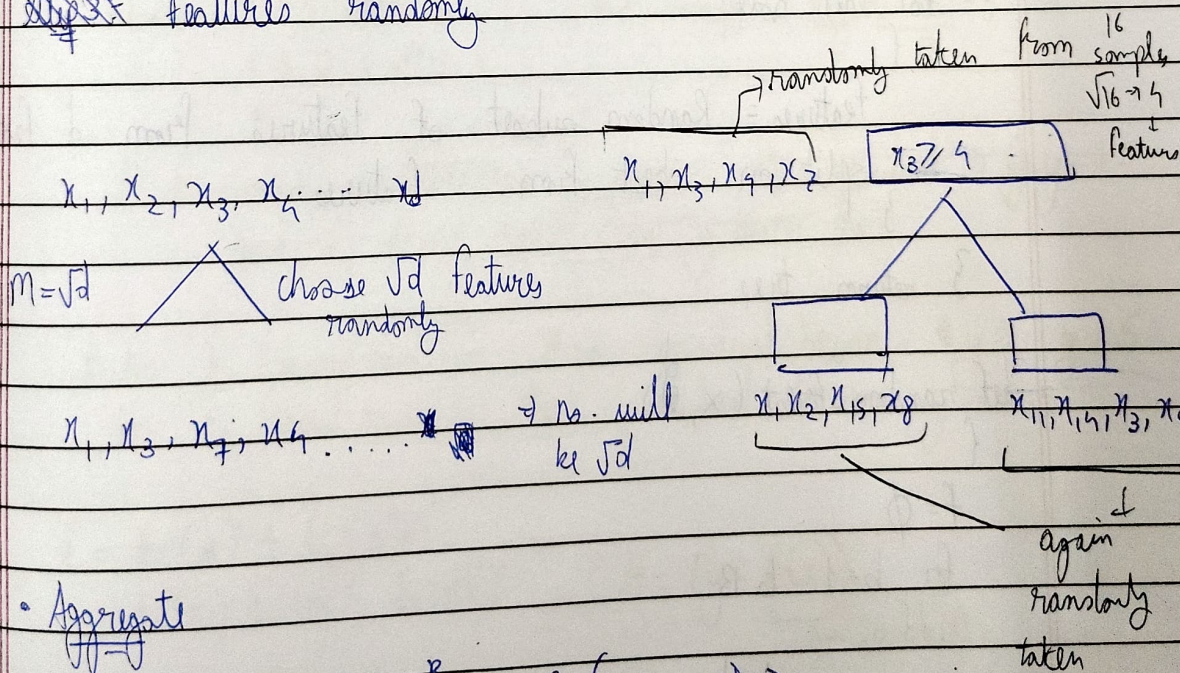


#B = no. of decision trees

• Step 1 :- generate multiple datasets from X [~~B~~ Bootstrapping]

• Step 2 :- Build a decision tree for each and every dataset
While building a decision tree, choose a subset of

• ~~Step 3~~ features randomly



• Aggregate

$$f(x; \theta) = \frac{1}{B} \sum_{b=1}^B f_b(x_b; \theta_b) \Rightarrow \text{regression}$$

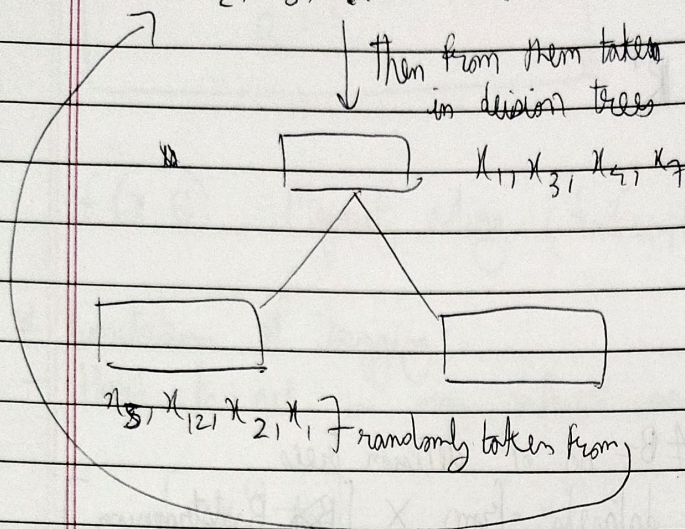
$$f(x; \theta) = \text{majority voting for classification setting.}$$

First the original dataset

↓ Randomly taken samples

$x_1, x_2, x_3, x_4, \dots, x_d$ # features

↓ then from them taken random features in decision trees



```
def Randomized-learn (X, d)
```

```
{  
  for each node
```

```
{  
    features = Random subset of features from d features  
    split on best from features
```

```
}  
return tree
```

```
def random-forest (X, d)
```

```
{  
  f =  $\phi$ 
```

```
  for b = 1 to B
```

```
  {  
     $x_b$  = A bootstrapped dataset from X
```

```
     $f_b$  = Randomized-learning learn ( $x_b, d$ )
```

```
     $f = f \cup \{f_b\}$ 
```

```
  }  
  ↵ union
```

```
  return f
```

```
}
```